# BRYAN PETRIE

SOFTWARE ENGINEERING & INFORMATION TECHNOLOGY

**My Website**: Bryan Petrie - Portfolio

## CONTACT

319-930-6072
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## SKILLS

- Python & Java
- SQL & Database Design
- Web Development (HTML/CSS)
- System Administration
- PowerShell Scripting
- VM/Hypervisor Management

## EDUCATION

*William Penn University*
***2024-2027***
*-BS in Software Engineering and Information Technology*
*-Minor in Business Management*
*(3.93 GPA)*
*-Dual sport athlete and a captain in soccer and track while completing studies*

## PROFILE

Software Engineering student with hands-on experience in full-stack development, system administration, and database management. Proven ability to develop user-friendly applications and solve complex technical problems. Strong communicator with experience training others and managing technical projects from concept to deployment.

## EXPERIENCE

**Line Cook at Buff's Gameday**
*Sept. 2022- Jan. 2026*
Gained experience in a work establishment and developed skills in communication and work area management. Assisted in training new employees and establishing kitchen standards while maintaining health standards.

**William Penn University Lab Assistant**
*2025-2027*
Managed and assisted over 300 students with software and programming issues by guiding them through the processes so they understood how to navigate the problems themselves.

**Undergraduate Researcher, NSF REU — ASTAR Lab, University of Nebraska Omaha**
*Summer 2026*
Developed an adversarial attack pipeline targeting cross-view geo-localization models, which predict where a photograph was taken by matching it against satellite imagery. Built and iteratively refined the attack in PyTorch, substantially improving its success rate while keeping image perturbations imperceptible, and evaluated it against multiple model architectures to compare their adversarial robustness. Analyzed which images resisted attack and why, then authored a first-author research paper on the findings under the IARPA WRIVA Challenge.

**Undergraduate Research**

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- Designed and implemented a targeted adversarial attack pipeline against cross-view geo-localization retrieval models, raising attack success from 26.2% to 78.2% at a fixed perturbation budget through embedding-cache target selection, momentum-accumulated PGD, dynamic retargeting, and a two-stage execution strategy
- Identified that adversarial vulnerability is a property of the source scene rather than the source-target pairing via a controlled substitution experiment, and showed that resistance correlates with input-gradient magnitude rather than any image-level visual statistic
- Extended the attack to a DINOv3 vision-transformer CVGL architecture, measuring a 30-point robustness gap against the convolutional baseline
- Authored a first-author IEEE-format research paper on the findings; work conducted under the IARPA WRIVA Challenge

**PROJECTS**

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**AI CHATBOT DEVELOPMENT**
Developed, trained using LLM and large-scale databases, and implemented chatbots for different local businesses to aid in customer support and employee onboarding.

**POS DEVELOPMENT**
Assisted in the implementation and development of a "Point of Service" system in a local establishment. Consulted staff to ensure the system was user friendly and flexible.

**TASK MANAGER DEVELOPMENT**
Programmed and deployed a personalized task manager onto PC. Used to not only manage tasks but increase productivity and performance of the user and machine. Also used to help manage multiple VM's at a time.

## TECHNICAL SKILLS

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**PROGRAMMING LANGUAGES**
Python: Data processing, automation scripts, and application development
Java: Object-oriented design, software development, and debugging
SQL: Database queries, optimization, and relational database design
HTML/CSS: Responsive web design and user interface development
PowerShell: System automation and administrative task scripting

**SYSTEMS & INFRASTRUCTURE**
Virtual Machine Management: Creation, deployment, configuration, and maintenance of VMs
Network Administration: Configuration, troubleshooting, and security implementation
System Administration: Server management, user access control, and system monitoring
Database Management: Schema design, data modeling, query optimization, and backup procedures

**MICROSOFT AZURE CLOUD SERVICES**
Experience with Azure cloud infrastructure: VM deployment and management, Active Directory for identity and SSO, SQL Database administration and query optimization, resource group organization, CI/CD pipelines in Azure DevOps, and Virtual Network configuration with subnet segmentation and security controls.